# Hyperspectral Object Tracking With Context-Aware Learning and Category Consistency

Ye Wang, Shaohui Mei, *Senior Member, IEEE*, Mingyang Ma, Yuheng Liu, Tao Gao, *Member, IEEE*, and Huiyang Han

***Abstract*— Hyperspectral imaging technology is of crucial importance to improve the performance of object tracking in many remote sensing surveillance areas. Previous methods primarily focused on feature fusion strategies by employing additional enhancement modules. However, these methods commonly lack contextual understanding to distinguish the target from the background and totally ignore the category information of the targets. To address these limitations, a novel hyperspectral object tracker is proposed to incorporate context-aware learning and category consistency tracker (CCTrack), which can adaptively learn context-aware representations in hyperspectral scenarios to obtain global target information with memory storage, while constructing an interframe category consistency constraint to enhance tracking process. Specifically, CCTrack integrates an adaptive context-aware learning (ACL) mechanism, which includes a feature decoupling module (FDM) to extract specific representations from decoupled features, and a Mamba layer to retain and update long-range dependencies. To align with prior knowledge of target recognition and motion patterns, an alignment transformation module (ATM) is employed with the ACL mechanism, fully leveraging spatial–spectral representations. In addition, category consistency constraint modules (C3Ms) are introduced to enforce category consistency across frames by computing the similarities between the target features and the corresponding category name, serving as the constraint to improve tracking performance. Extensive experiments over the hyperspectral object tracking (HOT) benchmark covering various remote sensing scenarios demonstrate that CCTrack outperforms state-of-the-art methods by a significant margin.**

***Index Terms*— Category consistency, context-aware learning, hyperspectral imaging, object tracking, remote sensing.**

## I. Introduction

REMOTE sensing has long been utilized to gather data about distant objects and phenomena, primarily through satellite and airborne sensors for broad-scale environmental monitoring, such as land cover classification [1], vegetation analysis [2], [3], and atmospheric studies [4], [5], [6]. In recent years, hyperspectral imaging has emerged as a more advanced form of remote sensing, capturing data across a wide range of wavelengths [7], [8] and allowing for the identification and differentiation of materials and objects based on spectral signatures, with researched methodologies including spectral unmixing [9], target detection [10], super-resolution [11], [12], and classification [13], [14], [15]. Hyperspectral object tracking (HOT), which uses hyperspectral imagery for identifying and tracking objects like vehicles or aircraft, takes remote sensing a step further by enabling real-time monitoring of dynamic objects across vast areas. By capturing detailed spectral information, hyperspectral tracking systems can distinguish objects even in cluttered environments or challenging conditions, such as varying lighting, complex background, or camouflage [16], [17], [18].

The high dimensionality of hyperspectral data, however, poses a significant challenge for HOT, as it requires processing and analyzing large amounts of data in real-time. Moreover, the limited sample scale in hyperspectral data makes it difficult to learn robust representations for tracking objects across frames. Initial HOT methods focused on hand-crafted features, such as spatial–spectral histograms of multidimensional gradients with material abundances [16], histograms of oriented mosaic gradients [19], and reformulated stacked histograms [20]. In these methods, correlation filters (CFs) are used to learn the target appearance model and predict the target location in the next frame. However, the hidden target representations and complex motion patterns in hyperspectral data are difficult to capture with hand-crafted features and CF-based models, leading to performance degradation in challenging tracking scenarios.

Inspired by the success of deep learning in computer vision tasks [21], [22], [23], recent HOT methods have shifted toward deep learning-based approaches, which can automatically learn spatial–spectral representations from hyperspectral data. These methods typically employ deep convolutional neural networks (CNNs) or transformers to extract hyperspectral features and integrate with additional enhancement modules. For example, Liu et al. [24] and Wang et al. [25] proposed a false color feature extraction module to obtain spatial information from hyperspectral data and enhance the spectral representation with additional hyperspectral feature fusion. Li et al. [18] learned a band importance map to adaptively select informative bands and ensemble multiple CNNs for spatial–spectral feature fusion. Zhao et al. [26] further introduced the RGB data as

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an auxiliary input to improve the spatial–spectral feature learning, achieving multimodal feature fusion for HOT. These methods all involved the recognition that RGB data helps with the model's abilities on discriminating target and background and learning a likely motion patterns, and extracting features of hyperspectral data with pretrained extractor can produce an improved performance. However, they often ignore three critical aspects of hyperspectral tracking: 1) directly modeling spatial–spectral features and motion patterns in the hyperspectral data through an RGB-like feature extraction module may impair the hidden representations in particular contexts; 2) employing enhancement module to obtain separate enhanced spectral features may hinder the complex inner spatial–spectral correlations in target tracking process; and 3) though RGB data helps with strong prior, specific designs are required to ensure the reasonable combination of prior knowledge and hyperspectral context. Though some methods have attempted to address these issues by modifying the inner structure of the network to introduce prior knowledge [27] for limited data scale learning, the tracking performance is still suboptimal due to the lack of a robust context-aware learning in hyperspectral context. In addition, the category consistency across frames is often ignored in existing methods, which abandons the potential of exploiting the category information for tracking performance improvement.

To address these limitations, a novel HOT framework, category consistency tracker (CCTrack), is proposed to introduce an adaptive context-aware learning (ACL) mechanism and a category consistency constraint module (C3M). This framework fully explores the unique contextual information inherent in hyperspectral data by leveraging robust object motion priors and utilizing category-based auxiliary information to enhance tracking performance. Specifically, following previous works [18], [24], [25], [26], a transformer-based tracking model is first employed to learn robust motion patterns on readily available RGB tracking data. Instead of directly fine-tuning a pretrained deep network, the ACL mechanism is incorporated with an alignment transformation module (ATM) in CCTrack to decouple contextual information during both prior learning and spatial–spectral modeling process. The ACL mechanism retains strong identification and localization capabilities while capturing intrinsic contextual spatial–spectral representations in hyperspectral data. In addition, a vision-language foundation model is further employed in a C3M to enforce category consistency across frames, providing additional constraints that improve tracking performance. The primary contributions of this work are summarized as follows.

1) A novel HOT framework, CCTrack, is proposed, integrating a context-aware learning mechanism and a C3M into a unified model, which extracts and stores contextual hyperspectral representations and leverages category information to enhance the tracking process.
2) An ACL mechanism is introduced to learn contextual information from decoupled features and store long-range dependencies, effectively capturing the hidden target representations and complex motion patterns.
3) A C3M is employed to enforce category consistency across frames by computing similarities between visual features and corresponding category information, further enhancing tracking stability and accuracy.

The remainder of this article is as follows. Section II provides an overview of the related work. Section III describes the architecture of the proposed CCTrack in detail, with emphasis on the ATM, the ACL mechanism, and the C3M. Section IV presents extensive experimental results to demonstrate the effectiveness of the proposed approach. Finally, Section V summarizes the contributions of this work and explores possible directions for future research.

## II. Related Works

### A. Hyperspectral Object Tracking

HOT refers to the task of tracking a specific object in hyperspectral data sequences across frames, which is crucial for surveillance and remote sensing applications. HOT methods can be broadly categorized into two groups: traditional methods and deep learning-based methods. Traditional HOT methods primarily relying on designing hand-crafted spatial–spectral features specific to hyperspectral data, such as material abundances [16], oriented mosaic gradients [19], and reformulated stacked histograms [20]. After the feature extraction, the widely used CFs are employed to predict the target motions in each frame. These methods have shown promising results in HOT tasks, but require complicated feature engineering and are limited in capturing complex motion patterns in hyperspectral data.

However, deep learning-based HOT methods aim to automatically capture spatial–spectral features from hyperspectral data and have shown improved tracking performance compared to traditional methods. Compared to tracking objects in RGB data, HOT methods require to consider the high dimensionality of hyperspectral data and the limited sample scale, making it difficult to directly apply existing deep learning-based tracking models. It is realized that the easily accessible large-amount RGB datasets (e.g., GOT-10k [28]) can help learn a classification and motion prior for robust hyperspectral tracking. Current deep learning-based HOT methods typically employ a hybrid architecture, utilizing a prior-based spatial branch to extract appearance and positional variations, and a spectral branch to obtain enhanced spectral information. For example, Liu et al. [24] and Wang et al. [25] employed RGB-based Siamese networks and false color transformer as the main branch for feature extraction, respectively. As enhanced spectral acquisition, these two methods further introduced additional hyperspectral branches to fuse spatial–spectral features for HOT. Similarly, Li et al. [18] and Zhao et al. [26] also leveraged the RGB-based Siamese networks as the main branch for spatial feature extraction and introduced the spectral information through additional spectral perception designs. These methods have shown improved tracking performance compared to traditional methods, but the inherent contextual information in hyperspectral data may not be obtained by these feature fusion strategies.

### B. Context-Aware Learning

Context-aware learning is a broad concept that refers to the ability of a model to adaptively learn specific contextual

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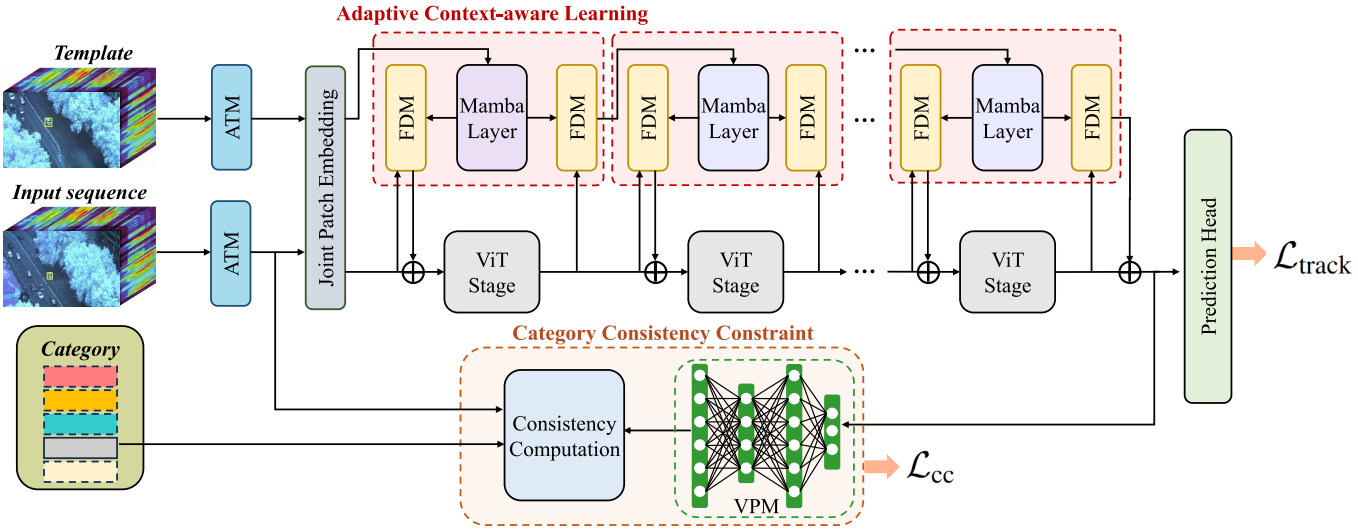



Fig. 1. Framework of CCTrack. The hyperspectral data is first aligned with the ATM and then fed into the transformer network with the ACL mechanism. The C3M is employed to enforce category consistency across frames with consistency computation and VPM.

information from the given data [29], [30], [31]. In the context of HOT, context-aware learning aims to capture the hidden spatial–spectral representations and complex motion patterns in hyperspectral data. Previous HOT methods mainly adopt ordinary RGB-based Siamese networks pretrained on RGB tracking datasets [18], [25], [26] as fundamental feature extractors and motion predictors, while incorporating additional hyperspectral branches to enhance the spectral information. However, these approaches may ignore the active learning of specific contextual information in hyperspectral data while maintaining the motion prediction abilities learned from RGB data, leading to the unbalanced spatial–spectral representations and suboptimal tracking performance. Inspired by the success of injecting adapters into the network to adaptively learn specific contextual information [32], [33], a novel ACL mechanism is proposed in this article to learn specific spatial–spectral representations in a hyperspectral context.

### C. Vision-Language Tracking

Vision-language tracking is a recent research direction that aims to integrate vision and language information for object tracking tasks [34], [35], [36]. The vision-language tracking methods typically utilize pretrained vision-language models (e.g., CLIP [37] and ViLBERT [38]) to learn the joint representations of vision and language information for tracking tasks. However, these methods require carefully labeled language descriptions for each frame, which may not be available in HOT scenarios. Another language-assisted visual tracking method is to transform text description into discrete attributes [39], e.g., using "color" and "texture" to identify the target. However, the description of this attribute can be inaccurate between pretrained domain and the hyperspectral domain, which is not reliable. In this work, instead of achieving vision-language multimodal tracking and using attributes for tracking, a C3M is introduced to improve the training stability of the proposed method by only utilizing the available category information across frames.

## III. Method

### A. Overall Framework

The overall framework of the proposed CCTrack is illustrated in Fig. 1. Given a hyperspectral sequence, the proposed CCTrack first extracts a search region corresponding to the template in the first frame. The search region and the template are then transformed into an informative hyperspectral space by an ATM for the alignment of the target recognition and motion prior. The spectral-wise aligned features are then fed into a transformer-based tracking model for feature extraction. Meanwhile, an ACL mechanism is employed, utilizing ACL blocks to adaptively learn the specific spatial–spectral representations in a hyperspectral context. In addition, a C3M is proposed to employ the CLIP model to enforce category consistency across frames in the training process. The final tracking results are obtained by the subsequent predictive heads based on the learned spatial–spectral representations.

### B. Alignment Transformation Module

Unlike previous methods utilizing band selection to choose three bands as false color images for feature extraction [24], [25], the ATM employs a spectral attention mechanism to achieve this alignment, as shown in Fig. 2. This transformation not only eliminates the need for an additional spectral enhancement module but also facilitates ACL in subsequent components, enhancing the model's ability to extract meaningful spectral features effectively.

Specifically, given a search region $\mathcal{S} \in \mathbb{R}^{W\times H\times N}$ in a single frame $\mathcal{X}$ from the hyperspectral sequence, $N$ is its number of bands with spatial resolution of $W \times H$. It is first projected into a higher dimensional space by a linear mapping as feature embedding

$$\mathcal{F}_{\mathcal{S}} = \text{norm}(\text{Linear}(\mathcal{S})) \tag{1}$$

where norm$(\cdot)$ denotes the normalization operation (e.g., batch normalization). The feature embedding is then fed into a

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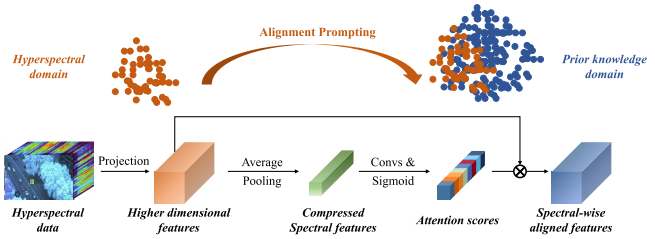



Fig. 2. Structure of the ATM. The hyperspectral data is projected into a higher dimensional space and then fed into a spectral attention module to obtain the spectral attention weights.

spectral attention module to obtain the attention weights, which can be used to obtain spectral-wise aligned features

$$\mathcal{F}_{\text{spec}} = \text{Sigmoid}(\text{CR}(\text{CR}(\text{AvgPool}(\mathcal{F}_{\mathcal{S}})))) \odot \mathcal{F}_{\mathcal{S}} \quad (2)$$

where CR denotes the ordinary convolution and ReLU operations, and $\odot$ denotes element-wise multiplication. The meaningful spectral information is thus effectively extracted through the spectral attention mechanism, which enhances the most informative spectral features while suppressing less relevant ones. This refined spectral representation is then projected back to a lower dimensional space, reducing computational complexity while preserving critical information for the subsequent tracking model. Furthermore, the ATM plays a crucial role in mitigating the domain gap between prior knowledge and hyperspectral data. Traditional tracking models often struggle with hyperspectral information due to its high dimensionality and unique spectral distribution. However, by leveraging spectral attention, the ATM enables the hyperspectral data to adaptively align with the prior knowledge domain, facilitating context-aware learning. This alignment ensures that hyperspectral features are effectively utilized within the designed tracking mechanisms, leading to enhanced tracking performance in complex scenarios.

### C. Adaptive Context-Aware Learning

ACL mechanism is embedded into the transformer feature extraction network to learn specific hyperspectral representations, as shown in Fig. 1. To obtain hierarchical spatial–spectral features, the transformer network is evenly divided into four stages, each of which is equipped with an ACL block. Instead of employing an enhancement module for additional spectral information learning, the ACL mechanism aims to decouple spatial and spectral features from main stream flow, thus achieving contextual perception and flexible active learning. The detailed structure of the ACL block is illustrated in Fig. 3. The ACL block consists of two feature decoupling modules (FDMs) and a dynamic selectively updating Mamba layer, which is designed to decouple prior feature and hyperspectral feature representations. This structure enables the model to learn task-specific feature representations dynamically, reducing feature entanglement and enhancing tracking robustness in hyperspectral scenarios. The context-aware learning is achieved in a two-step process: 1) the foreground and background discrimination and motion

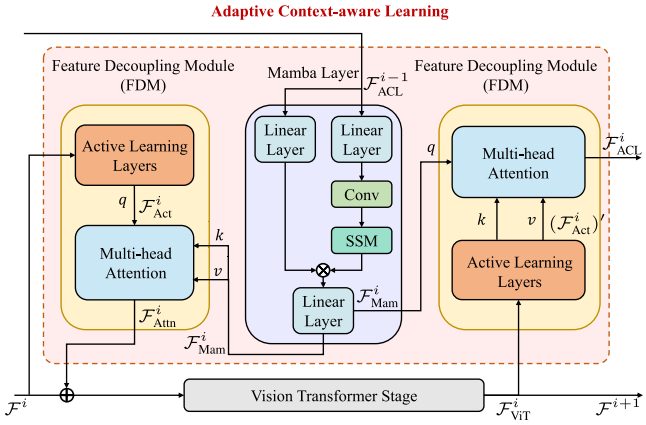



Fig. 3. Structure of the ACL block. The ACL block consists of two FDM, two MA layers, and a dynamic selectively updated Mamba layer.

patterns learning and 2) the hyperspectral-specific contextual target representations learning.

In the first step, active learning layers in the FDM are deactivated, and multihead attention (MA) layers and the updating Mamba layer are integrated into the transformer network. The tracker is then trained on the easily accessible RGB dataset, following the common protocol. Though the transformer network itself can extract the discriminative features for the target recognition and localization, the representative features are deeply coupled in the feature flow, which causes the difficulty in learning subsequent specific spatial–spectral representations. To decouple the feature flow for better hyperspectral information perception, each ViT stage with an ACL is employed to analyze the deconstructed features. The feature decoupling mechanism ensures that the most informative spectral characteristics are preserved to enhance the learned representations. To complement the MA layers, an updating Mamba layer is also employed. Mamba [40] is a variation of the state space model (SSM) designed to store global contextual information as memory states while dynamically updating features. In the ACL block, the updating Mamba layer is incorporated to capture long-term dependencies by selectively compressing the learned global context information and refining the target representations. Unlike conventional transformers, the updating Mamba layer efficiently maintains and adapts feature representations over time through adaptively parameter adjustments. In addition, due to its efficient global perception ability, it serves as an information supplement for long-distance representation modeling at different levels. By integrating the MA layers with the updating Mamba layer, the ACL block effectively utilizes the deconstructed features, while adaptive learning enables the acquisition of new knowledge based on prior knowledge. This combination enables the model to dynamically refine hyperspectral feature representations while maintaining efficiency. Note that the ATM is also removed in this stage due to no alignment requirement.

In the second step, the active learning layers are activated to perceive fine-grained contextual differences between the prior spatial and motion knowledge and the unique hyperspectral representations. During this process, the primary feature flow

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established by the transformer network remains stable to ensure accurate target recognition and localization, while the active learning layers focus on facilitating the convergence of hyperspectral-specific contextual tracking. The ATM and ACL are collaboratively designed to adaptively learn complex spatial–spectral representations and motion patterns in hyperspectral data, which balances the prior knowledge and learning the unique contextual characteristics of hyperspectral scenarios.

Specifically, for the input feature map $\mathcal{F}^i \in \mathbb{R}^{(W'\times H')\times D}$ before the $i$th transformer stage, $D$ is the embedding dimension and $(W' \times H')$ is the reformulated spatial feature resolution. It is fed into the active learning layers, which is a three-layer MLP, for fine-grained decoupled features learning

$$\mathcal{F}^i_{\text{Act}} = \text{MLP}\left(\mathcal{F}^i\right). \tag{3}$$

The Mamba layer receives the last ACL block's output represented by $\mathcal{F}^{i-1}_{\text{ACL}}$ for long-term dependency injection

$$\mathcal{F}^i_{\text{Mam}} = \text{Mamba}\left(\mathcal{F}^{i-1}_{\text{ACL}}\right) \tag{4}$$

where the inner structure of Mamba layer is shown in Fig. 3. The critical component in the Mamba layer is the SSM, which predicts the output $y_t$ based both on the input $x_t$ and the hidden state $h_t$

$$\begin{aligned} h_t &= \mathbf{A}h_{t-1} + \mathbf{B}x_t \\ y_t &= \mathbf{C}h_t \end{aligned} \tag{5}$$

where $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are the transition matrices. The SSM helps with retaining the global information learned in different contexts while updating the new knowledge, which is effective with MA for feature decoupling. Instead of using constant transition matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, the SSM in Mamba layer using dynamic matrices determined by the input $x_t$ through linear projection functions regard to $x_t$. By this dynamic transition matrix design, the Mamba layer can selectively focus on the previous state or new information. After obtaining $\mathcal{F}^i_{\text{Act}}$ and $\mathcal{F}^i_{\text{Mam}}$, the MA layers are employed to promote the global contextual information interactions and form a decoupled feature flow

$$\mathcal{F}^i_{\text{Attn}} = \mathcal{F}^i_{\text{Act}} + \alpha^i \,\text{Attn}\left(\text{LN}\left(\mathcal{F}^i_{\text{Act}}\right), \text{LN}\left(\mathcal{F}^i_{\text{Mam}}\right)\right) \tag{6}$$

where $\text{Attn}(\cdot)$ denotes the MA layers, $\text{LN}(\cdot)$ denotes the layer normalization, and $\alpha^i$ denotes the learning parameter to balance the two features. The attention feature map is then fused with the original input feature and passed through a normal transformer stage

$$\mathcal{F}^i_{\text{ViT}} = \text{ViT}\left(\mathcal{F}^i_{\text{Attn}} + \mathcal{F}^i\right) \tag{7}$$

where $\text{ViT}(\cdot)$ denotes a vision transformer stage which contains $L/4$ transformer layers, and $L$ is the total number of transformer layers. Note that $\mathcal{F}^i_{\text{ViT}}$ is also the input feature of the next stage $\mathcal{F}^{i+1}$. After output by the ViT stage, another FDM is employed to achieve postdecoupling and feature propagation, also fusing the Mamba feature

$$\left(\mathcal{F}^i_{\text{Act}}\right)' = \text{MLP}\left(\mathcal{F}^i_{\text{ViT}}\right) \tag{8}$$

$$\mathcal{F}^i_{\text{ACL}} = \mathcal{F}^i_{\text{Mam}} + \beta^i \,\text{Attn}\left(\text{LN}\left(\mathcal{F}^i_{\text{Mam}}\right), \text{LN}\left(\left(\mathcal{F}^i_{\text{Act}}\right)'\right)\right) \tag{9}$$

where $\mathcal{F}^i_{\text{ACL}}$ is the output feature of the $i$th ACL block and the input of the Mamba layer in the next block, and $\beta^i$ is also the learnable balancing parameter. The two active learning layers are replaced with identify mapping in the first step.

### D. Category Consistency Constraint Module

In single object tracking tasks including HOT, the initial ground-truth bounding box is commonly regarded as the primary reference for matching targets across consecutive frames. However, the tracking model may occasionally lose the target due to factors such as occlusion (OCC), illumination changes, or other challenging conditions. In this case, category information can serve as an additional cue, offering confidence references that reinforce the consistency of the target object across frames.

To address this, the C3M is introduced to help the proposed CCTrack learn more stable and consistent tracking representations by enforcing category consistency across frames, employing the CLIP model [37] as the category auxiliary to provide additional category prompts for the training process. The CLIP model is pretrained on a large-scale vision-language dataset, which contains a wide range of category information, providing a robust object-category prior to category constraint modeling.

As shown in Fig. 4, in the training process, the category information is projected into text embeddings, and concatenated with the projected embedded features from the main feature stream. The embedded features are projected by a visual projection module (VPM), implemented as a two-layer MLP, to generate context-aware features that enhance the category information prior. To further refine the projected features, learnable tokens are introduced. These tokens provide additional flexibility, allowing the model to dynamically adjust feature representations. The concatenated features, enriched with both category and context-aware information, are subsequently encoded by the text encoder of the CLIP model. Simultaneously, the image encoder processes the spectral-wise aligned features, generating corresponding image features. Finally, similarity scores are computed by comparing the encoded text and image features, facilitating more robust and semantically meaningful feature matching. The category consistency loss is calculated by the cosine similarity between the image–text similarity scores, which is further utilized to enforce the category consistency across frames. The VPM is implemented as a two-layer MLP, which is employed for the alignment of the hyperspectral visual representations and the pretrained text information.

Specifically, for the spectral-wise aligned feature $\mathcal{F}_{\text{spec}} \in \mathbb{R}^{W\times H\times 3}$, it is fed into the image encoder $\text{Im}(\cdot)$ for image features

$$\mathcal{F}_{\text{im}} = \text{Im}(\mathcal{F}_{\text{spec}}). \tag{10}$$

The text features are generated by passing the concatenation features of text embedding and projected embedded features to the text encoder

$$\mathcal{F}_{\text{tx}} = \text{Tx}(\mathcal{F}_{\text{concat}}) \tag{11}$$

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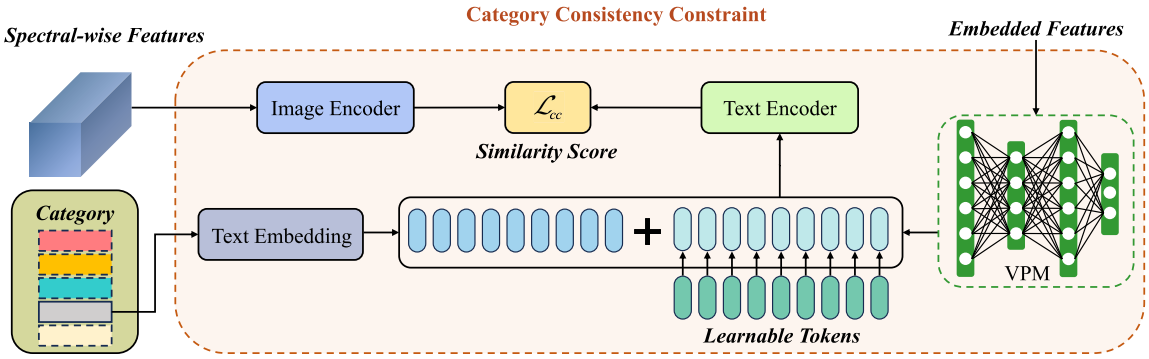



Fig. 4. Structure of the C3M.

where $\mathrm{Tx}(\cdot)$ is the text encoder of CLIP, and $\mathcal{F}_{\text{concat}}$ can be formulated as

$$\mathcal{F}_{\text{concat}} = \text{Concat}(\text{Tokenize}(\mathcal{N})\text{MLP}(\mathcal{F}_{\text{emb}}) + \mathcal{T}) \tag{12}$$

where $\text{Concat}(\cdot)$ denotes the concatenation operation, $\text{Tokenize}(\cdot)$ denotes the tokenization in CLIP, $\text{MLP}(\cdot)$ is the VPM, $\mathcal{N}$ is the input category name, $\mathcal{F}_{\text{emb}}$ denotes the embedd features, and $\mathcal{T}$ denotes the learnable tokens. It is notable that, the category name in the hyperspectral dataset is not strictly aligned with its content. Therefore, the category name correction is performed to align the category name, e.g., changing "doublecar2" to "car" for the right target.

For final tracking results, three predictive heads are employed to localize the target object by predicting the classification score map $\mathcal{C}$, the bounding box size $\mathcal{S}$, and the bounding box offset $\mathcal{O}$, respectively, following [41]. Supposing the prediction $(x, y, w, h)$ denotes the final $x$- and $y$-coordinates of the bounding box's center and its corresponding weight and height, it can be calculated by

$$(x, y, w, h) = (x_c + \mathcal{O}[x], y_c + \mathcal{O}[y], \mathcal{S}[x], \mathcal{S}[y]) \tag{13}$$

where $x_c$ and $y_c$ are most probable position in the classification map $\mathcal{C}$

$$(x_c, y_c) = \text{argmax}\ \mathcal{C}. \tag{14}$$

The tracking loss is formulated as the combination of three losses

$$\mathcal{L}_{\text{track}} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{size}}\mathcal{L}_{\text{size}} + \lambda_{\text{off}}\mathcal{L}_{\text{off}} \tag{15}$$

where $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{size}}$, and $\mathcal{L}_{\text{off}}$ denote the losses of classification score, bounding box size and bounding box offset, and $\lambda_{\text{cls}}$, $\lambda_{\text{size}}$, and $\lambda_{\text{off}}$ are the corresponding balance weight, which are set to 1, 5, and 2, respectively.

The total loss function of the proposed CCTrack is formulated as

$$\mathcal{L} = \mathcal{L}_{\text{track}} + \lambda_{\text{cc}}\mathcal{L}_{\text{cc}} \tag{16}$$

where $\mathcal{L}_{\text{track}}$ is the tracking loss, $\mathcal{L}_{\text{cc}}$ is the category consistency loss, and $\lambda_{\text{cc}}$ is the weight of the category consistency loss. For a more stable initialization, $\mathcal{L}_{\text{cc}}$ is set to be a small value at the beginning of the training process, and gradually increased as training progresses to enhance model performance.

## IV. Experiments

### A. Implementation Details

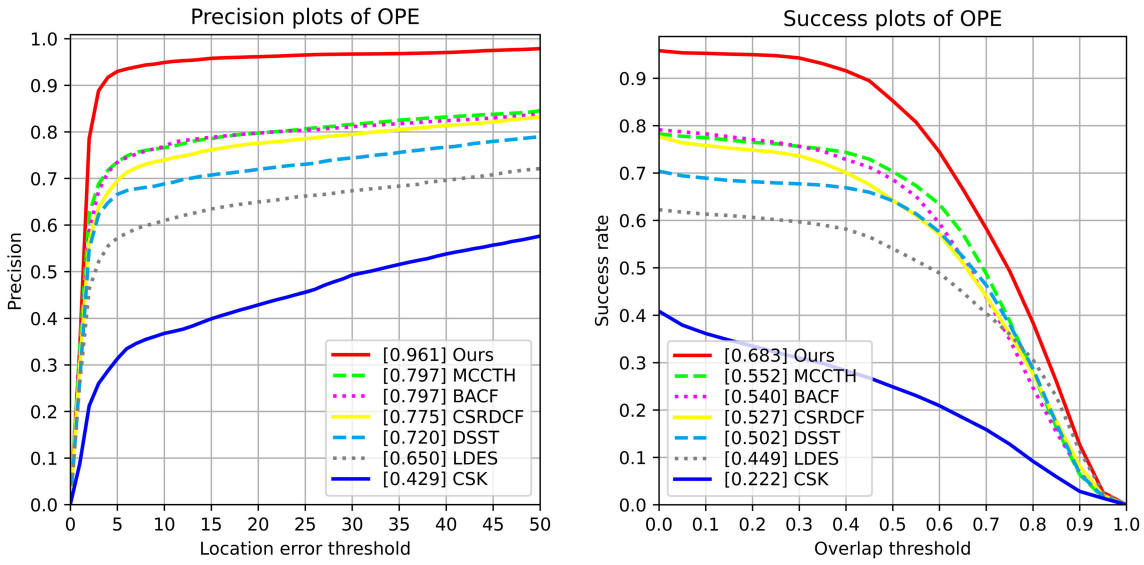
The training process is conducted on a single NVIDIA RTX 4090 GPU with 24-GB memory. The batch size is set to 8. The learning rate is initialized to $4 \times 10^{-4}$ and decayed by a factor of 0.1 and the total number of epochs is set to 40. The AdamW [42] optimizer is employed for optimization, and the weight decay is set to $1 \times 10^{-4}$.

### B. Dataset

The proposed CCTrack is first evaluated on the remote sensing-based HOT benchmark IMEC25 dataset [17], which contains 135 hyperspectral sequences with various challenging factors. The dataset covers a wide range of remote sensing scenarios, including surveillance and tracking of aviation, navigation, and transportation objects. The dataset is divided into a training set with 55 sequences of 14 060 frames and a testing set with 80 sequences of 14 107 frames. The hyperspectral data is captured across 25 bands ranging from 680 to 960 nm, with a spatial resolution of 409 × 216 pixels. By evaluating this HOT benchmark, the abilities of the proposed CCTrack in handling complex remote sensing scenarios and capturing specific spatial–spectral representations are fully demonstrated. In addition, the IMEC25 Dataset gathers the hyperspectral videos considering several challenging factors, such as background clutter (BC), illumination variation (IV), OCC, scale variation (SV), in-plane rotation (IPR), out-of-plane rotation (OPR), out-of-view (OV), and deformation (DEF). From these challenging factors, the tracking performance of the proposed CCTrack can be evaluated in a broad range of application scenarios.

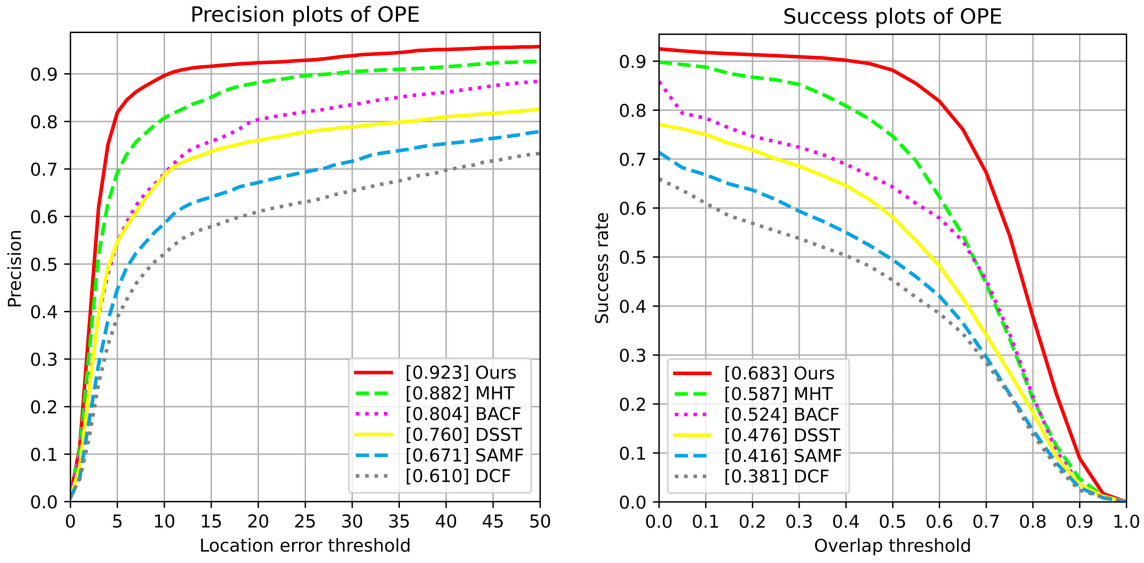
Meanwhile, CCTrack is also evaluated on another HOT benchmark IMEC16 dataset [16], which each frame has a resolution of 512 × 256 pixels and includes 16 spectral bands, spanning wavelengths from 470 to 620 nm. The dataset is divided into 40 training sequences containing 12 837 frames and 35 testing sequences with 16 574 frames. Similar to other object tracking datasets, each sequence in IMEC16 is annotated with a variety of challenges. These attributes provide a comprehensive framework for evaluating and tracking performance under diverse and challenging conditions.

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(a) (b)

Fig. 5. Overall tracking performance of the proposed CCTrack is compared with other CF-based methods on IMEC25. (a) Precision plots. (b) Success plots.




(a) (b)

Fig. 6. Overall tracking performance of the proposed CCTrack is compared with other CF-based methods on IMEC16. (a) Precision plots. (b) Success plots.

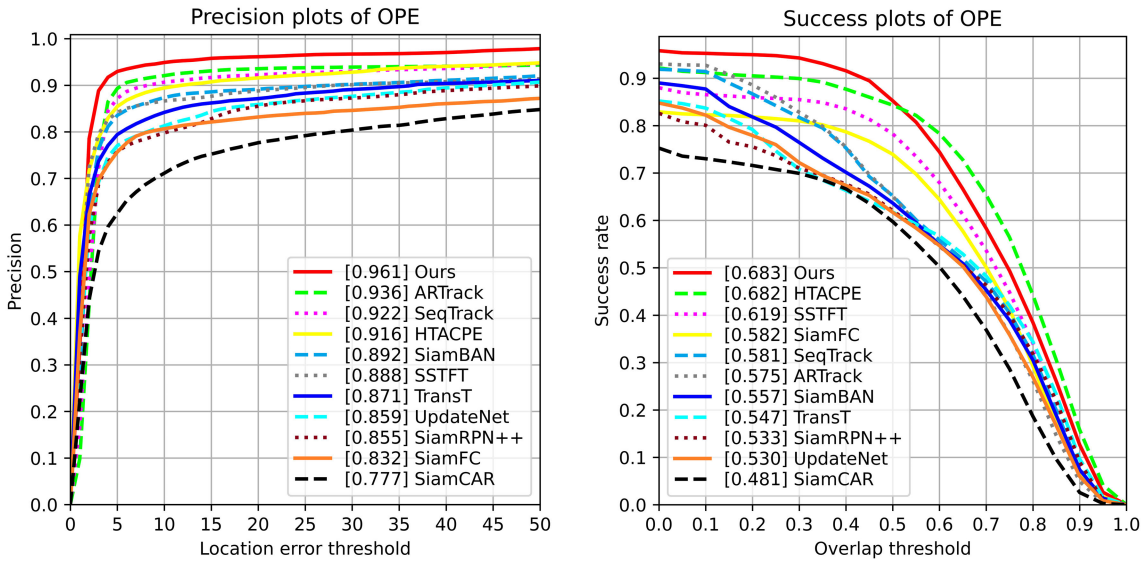
### C. Evaluation Metrics

The tracking performance is evaluated with the common one-pass evaluation (OPE) protocol, which calculates the success plots, the precision plots, the distance precision (DP), and the area under the curve (AUC) for each sequence. The success plots is defined as the percentage of frames whose intersection over union (IoU) between the predicted bounding box and the ground-truth bounding box is above a certain threshold. The precision plots is defined as the percentage of frames which DP between center points. The DP is defined as the percentage of frames whose predicted bounding box center is within a certain distance threshold from the ground-truth bounding box center, and the AUC is the area under the success plot curve with a threshold ranging from 0 to 1. The common distance threshold is set to 20 pixels to get the DP metric, and compute the area in the success plot is to get the AUC metric.

### D. Compared With State-of-the-Art Methods

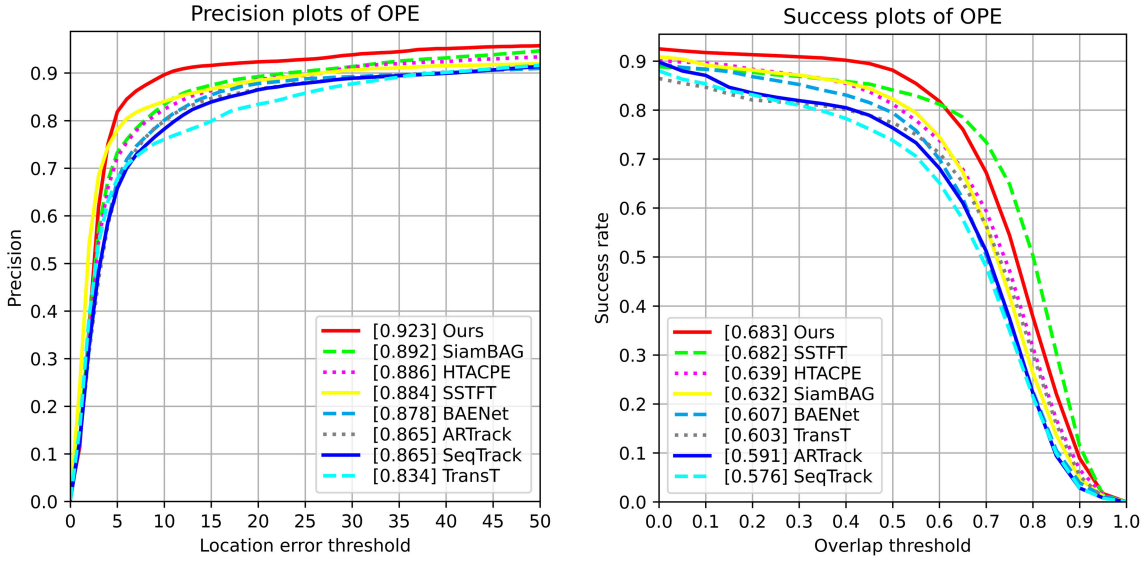
To demonstrate the effectiveness of the proposed CCTrack, several state-of-the-art single object tracking methods are compared on the HOT benchmark. The proposed method is compared in two public datasets IMEC25 and IMEC16. The overall tracking performance of the proposed CCTrack and the compared methods is shown in Figs. 5–8. It can be observed that the proposed CCTrack outperforms all the compared methods, achieving the best tracking performance in terms of both precision plots and success plots.

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(a) (b)

Fig. 7. Overall tracking performance of the proposed CCTrack is compared with other deep learning-based methods on IMEC25. (a) Precision plots. (b) Success plots.




(a) (b)

Fig. 8. Overall tracking performance of the proposed CCTrack is compared with other deep learning-based methods on IMEC16. (a) Precision plots. (b) Success plots.

*1) Compared With Filter-Based Methods:* As shown in Fig. 5, CCTrack is compared with state-of-art trackers including CSK [43], LDES [44], BACF [45], CSRDCF [46], and MCCTH [47] on IMEC25 dataset. Among all the compared methods, the hand-crafted filter-based methods all show inferior tracking performance owing to the limited feature representation in complex hyperspectral scenarios, thus affecting the motion prediction accuracy. Specifically, the efficient CSK obtains an unpromising performance with 22.2% in AUC and 42.9% in DP, which indicates it is challenging for the simplest circulant matrices to handle the high dimensional data and complex motions. Combined with more designed patterns, the filter-based methods can gain significant performance improvements. For example, BACF achieves 54% in AUC and 79.7% in DP and MCCTH achieves 55.2% in AUC and 79.7% in DP. However, the filter-based methods still lag behind the proposed CCTrack, with at least 13.1% and 16.4% performance gaps.

As shown in Fig. 6, CCTrack is compared with other state-of-art trackers including MHT [16], BACF [45], DSST [48], SAMF [49], and DCF [50] on IMEC16 dataset. Among all the compared trackers, MHT is the hyperspectral tracker. CCTrack demonstrates the best performance compared to these methods. The inferior performance of other trackers is attributed to their limited feature representation in complex hyperspectral scenarios, affecting tracking accuracy. Compared with the

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HOT method MHT, CCTrack achieves a 9.6% increase in AUC and a 4.1% improvement in DP. CCTrack also demonstrates excellent performance compared to other traditional trackers. For instance, when compared to BACF and DSST, CCTrack achieves improvements of 15.9% and 20.7% in AUC, and 12.8% and 16.3% in DP, respectively.

*2) Compared With Deep Learning-Based Methods:* As shown in Fig. 7, CCTrack is compared with other deep learning-based trackers deep learning-based methods, including SiamFC [21], SiamRPN++ [51], SiamCAR [52], SiamBAN [23], UpdateNet [53], TransT [54], SeqTrack [55], ARTrack [56], SSTFT [25], and HTACPE [27]. Deep learning-based methods generally show improved tracking performance compared to traditional methods. For example, even the worst performance tracker, SiamCAR, can achieving a DP of 77.7%, comparable with traditional hand-crafted methods. However, its AUC of 48.1% suggests that while it can coarsely perceive object motion, it struggles with accurate localization. Other Siamese-based deep learning methods achieve better results, with SiamFC achieves the best AUC of 58.2%, and SiamBAN get the highest DP of 89.2%. Though the Siamese-based methods are efficient and effective, they still lag behind the best performance methods due to the limited representation modeling. Among the transformer-based methods, TransT shows competitive tracking performance with Siamese series methods, inferior to other transformer-based methods. In contrast, the SeqTrack and ARTrack largely improve the performance by introducing sequential modeling mechanisms. However, though SeqTrack and ARTrack show high DP, the success rate is relatively low, which may indicate that they can robustly follow the trajectory thanks to the global understanding of the target-background relationship and motion patterns, but may not accurately localize the hyperspectral object due to the lack of specific spatial–spectral representations. The HOT-specific methods SSTFT and HTACPE, on the other hand, demonstrate high AUC but low DP, achieving 61.9% and 68.2% in AUC, and 88.8% and 91.6% in DP, respectively. This may be caused by the lack of robust motion prior and feature understandings in these two HOT methods. The proposed CCTrack, however, achieves the best tracking performance in both DP and a success rate of 68.3% in AUC and 96.1% in DP, which indicates the effectiveness of the proposed context-aware learning and C3Ms in capturing the hidden spatial–spectral representations as well as retaining the perception of the target inner features and motion patterns.

As shown in Fig. 8, the proposed Tracker CCTrack is compared against several deep learning-based methods, including SiamBAG [57], HTACPE [27], SSTFT [25], BAENet [58], ARTrack [56], SeqTrack [55], and TransT [54], on the IMEC16 dataset. The evaluation aims to highlight the strengths and limitations of existing object tracking approaches while demonstrating the improvements achieved by CCTrack. Among the compared methods, the CNN-based HOT method SiamBAG and BAENet achieve 63.2% and 60.7% in AUC, along with 89.2% and 87.8% in DP, respectively. Although both SiamBAG and BAENet leverage spectral information to a certain extent, their reliance on local feature extraction results in suboptimal tracking precision. The limited capability to capture global context and dynamic spectral variations leads to a decrease in tracking robustness, particularly in complex hyperspectral scenarios. On the other hand, transformer-based HOT methods HTACPE and SSTFT demonstrate higher AUC scores, achieving 63.9% and 68.2%, respectively. However, their DP scores remain lower, at 88.6% and 88.4%, compared to CNN-based trackers. This performance gap may stem from the fact that both HTACPE and SSTFT lack a category consistency constraint, which plays a crucial role in maintaining stable and accurate tracking over long sequences. While transformers are effective at capturing long-range dependencies, the absence of category information in these methods limits their ability to refine target representation, leading to reduced precision. In contrast, CCTrack outperforms these methods, demonstrating its superior ability to learn context-aware representations while integrating category consistency constraints to enhance tracking performance. Specifically, CCTrack achieves 5.1%, 0.1%, and 4.4% improvements in AUC and 3.1%, 3.9%, and 3.7% increases in DP, compared to SiamBAG, SSTFT, and HTACPE, respectively. These results highlight the effectiveness of CCTrack in leveraging both spatial–spectral contextual learning and category consistency, leading to more robust and precise tracking in hyperspectral scenarios.

To further demonstrate the tracking performance of the proposed CCTrack, several qualitative tracking results are visualized in Fig. 9. It can be observed that the proposed CCTrack can accurately localize the object in various challenging scenarios. For example, in the "airplane1" sequence, which includes only the LR attribute, the small-scale airplane is successfully tracked across the frames with low contrast and sensor motion. In contrast, existing methods such as SiamCAR and BACF fail to capture the object, while UpdateNet and SiamBAN exhibit inaccurate localization. Similarly, in the "boat" sequence, the proposed CCTrack can robustly track the boat object with severe oceanic background interference, while UpdateNet and BACF fail in the tracking process. In more complex scenarios, such as the "double5," which includes MB, OCC, FM, BC, LR, OPR, and DEF attributes, and "doublecar3" sequence, which includes OCC, BC, DEF, and IV attributes, all the compared methods fail to track the object. However, CCTrack accurately localizes the object across the frames, demonstrating its superior robustness in handling multiple simultaneous challenges. These qualitative results further demonstrate the effectiveness of the proposed CCTrack.

### *E. Attribute Analysis*

The performance of the proposed CCTrack in various challenging attribute scenarios is illustrated in Fig. 10 and compared with other state-of-art trackers. It is evident that CCTrack achieves the highest representative performance in several attributes, including SV, OCC, BC, DEF, OV, OPR, and IPR. These specific scenarios are chosen because they encompass the most common and critical challenges encountered in hyperspectral tracking, ensuring a comprehensive yet concise evaluation of our approach. Notably, CCTrack demonstrates

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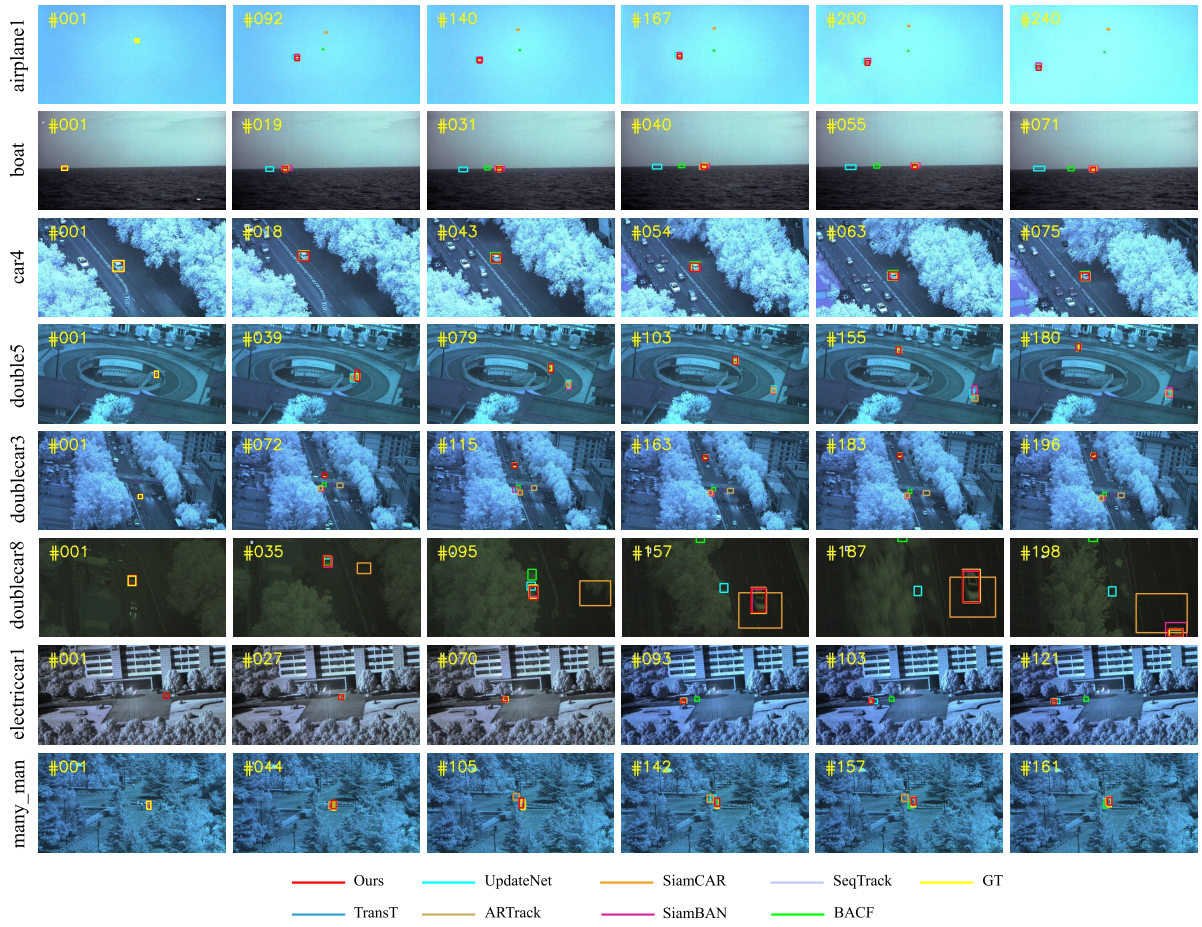



Fig. 9. Qualitative tracking results of the proposed CCTrack and the compared methods on the HOT benchmark.

strong robustness across various complex scenarios. In particular, CCTrack consistently outperforms the best performance by a significant margin in most attributes. CCTrack achieves a 2.8% improvement in AUC and a 2.4% improvement in the DP for SV compared to HTACPE and ARTrack, respectively. Furthermore, CCTrack surpasses SSTFT with a 5.9% improvement in AUC under OCC. In scenarios involving BC, CCTrack provides a 4.1% improvement in DP, while achieving a 2.6% improvement in DP under Def. For OV challenges, CCTrack shows a 4.8% improvement in AUC, and for OPR, it achieves a 5.1% improvement in AUC and 3.4% in DP. In addition, under IPR, CCTrack achieves substantial improvements, with a 9.2% increase in AUC and 7.7% in DP. Finally, despite achieving a smaller margin of improvement in IV, CCTrack still demonstrates competitive performance, with a 4.8% increase in AUC.

The detailed attribute analysis underscores the effectiveness of CCTrack in robustly identifying targets and modeling motion patterns across challenging scenarios. These results highlight the superior ability of CCTrack to leverage hyperspectral context, enabling it to address complex challenges with remarkable precision and adaptability. The strong performance across diverse attributes reflects the advanced design of CCTrack, particularly in integrating context-aware learning and category consistency modeling. Overall, the results confirm that the proposed CCTrack not only provides excellent target identification capabilities but also exhibits outstanding adaptability to hyperspectral-specific challenges. This makes CCTrack a robust and versatile solution for HOT in dynamic and complex environments.

### F. Ablation Study

To demonstrate the effectiveness of each component of CCTrack, ablation experiments are conducted, as presented in Table I. The contributions of the ATM, ACL block, and C3M are validated. Within the ACL block, the effectiveness of its subcomponents—the MA layers, the Mamba (Mam) layer, and the active learning (Act) layers are also discussed individually.

As shown in Table I, CCTrack achieves the highest AUC and DP scores across all experimental setups. Compared to the tracker without ATM, the full CCTrack achieves improvements of 1.5% and 1.9% in AUC and DP, respectively. Similarly, when C3M is excluded, the performance of CCTrack decreases by 0.9% and 1.2% in AUC and DP. In addition, further analysis highlights the importance of the three components within the ACL block. The MA layers serve as a foundational element in the ACL block, compared to a tracker that solely relies on the ATM, incorporating the MA layers leads to substantial performance gains, with an 8.5% improvement in AUC and a

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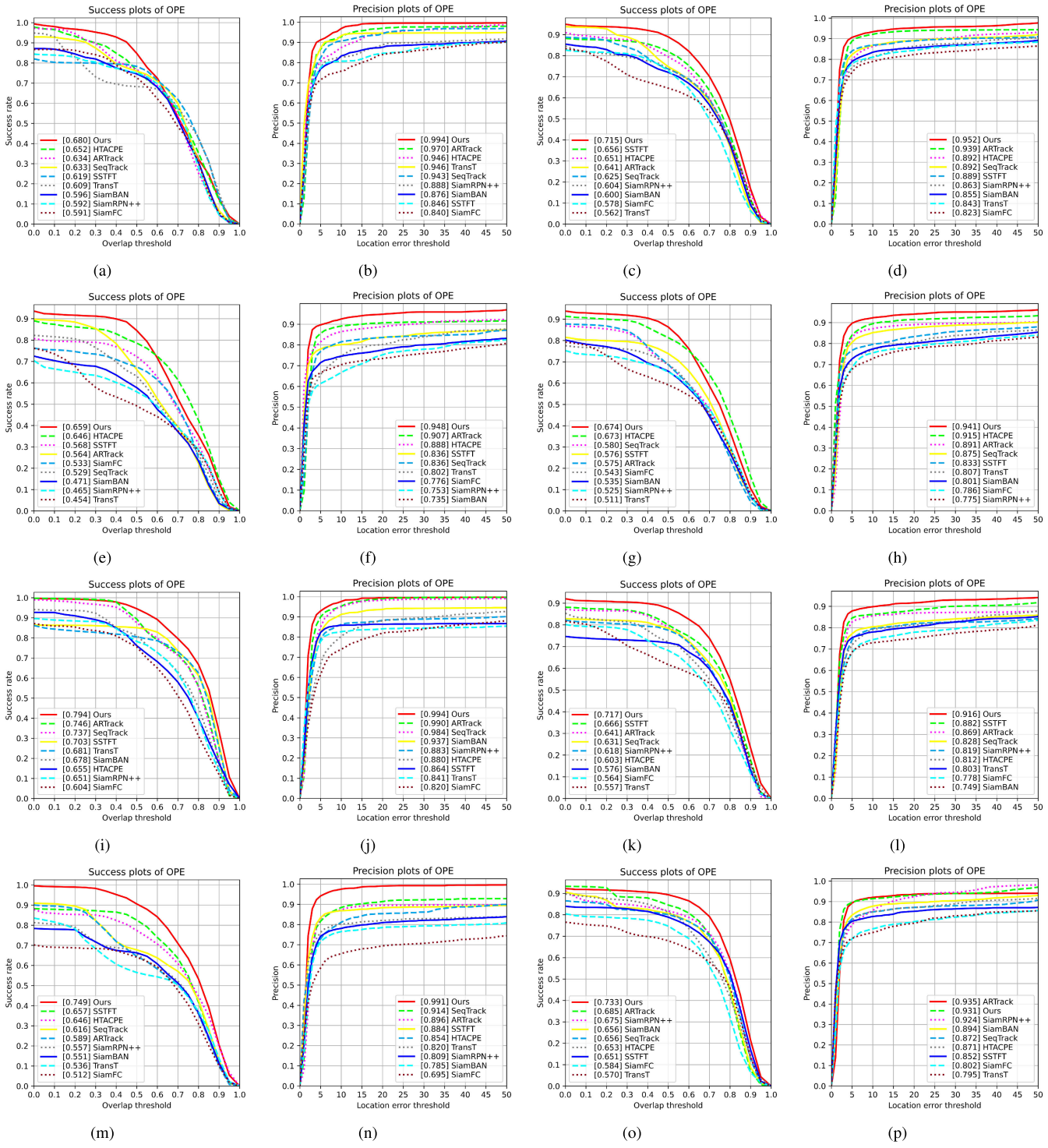



Fig. 10. Success and precision plot of compared trackers on IMEC25 with challenges. (a) and (b) SV. (c) and (d) OCC. (e) and (f) BC. (g) and (h) DEF. (i) and (j) OV. (k) and (l) OPR. (m) and (n) IPR. (o) and (p) IV.

7.2% increase in DP. These benefits can be further amplified by integrating the Mamba layer, which introduces efficient long-range dependency modeling and adaptive feature retention. When the Mamba layer is added to a tracker that already utilizes both ATM and MA, an additional performance boost of 2.8% in AUC and 2% in DP is observed, demonstrating its effectiveness in refining feature representation and improving tracking accuracy. By integrating the active learning layers, the ACL block becomes more effective to learn context-aware representations, resulting in further performance enhancements of 2.5% in AUC and 1.1% in DP.

Overall, the inclusion of ATM effectively aligns and processes the hyperspectral data, addressing the challenge of high-dimensionality and leading to a noticeable improvement in tracking performance. However, ATM alone does not provide a sufficient context-aware understanding of the

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TABLE I
ABLATION EXPERIMENTS ON EACH COMPONENT OF THE PROPOSED METHOD

| ATM | ACL | | | C3M | Evaluation Metrics | |
|---|---|---|---|---|---|---|
| | MA | Mam | Act | | AUC | DP@20 |
| | ✓ | ✓ | ✓ | ✓ | 0.668 | 0.942 |
| ✓ | | | | | 0.536 | 0.846 |
| ✓ | ✓ | | | | 0.621 | 0.918 |
| ✓ | ✓ | ✓ | | | 0.649 | 0.938 |
| ✓ | ✓ | ✓ | ✓ | | 0.674 | 0.949 |
| ✓ | ✓ | ✓ | ✓ | ✓ | **0.683** | **0.961** |

spatial–spectral features. The ACL block plays a crucial role in extracting contextual information from decoupled features, and its performance is enhanced by the integration of the MA layers, the Mam layer, and the Act layers. Finally, the inclusion of the C3M in the training process further enhances tracking accuracy by enforcing category-specific consistency across frames. The results of the ablation experiments clearly demonstrate that each component of CCTrack plays a significant role in addressing the challenges of HOT. The integration of these modules enables CCTrack to achieve state-of-the-art performance in terms of accuracy and robustness.

### G. Numbers of ACL

In the proposed method, the impact of the number of ACL blocks on model performance is explored in detail. The effect of varying the number of ACL blocks is analyzed in Fig. 11. The results demonstrate that utilizing an appropriate number of ACL blocks leads to optimal performance. When no ACL blocks are used (number set to 0), the tracking performance is the lowest among all configurations. As the number of ACL blocks increases to 1, 2, and 4, the tracking performance improves steadily. However, when the number is increased to 6, a slight decline in performance is observed.

In summary, increasing the number of ACL blocks enhances modeling abilities for the CCTrack to decouple features and model contextual information, thereby improving performance. However, excessively increasing the number of ACL blocks does not yield additional improvements and instead results in a slight performance drop. This decline is likely due to challenges in network optimization and potential inefficiencies introduced by overly complex configurations.

### H. Hidden State Dimension

In the Mamba layer, the dimension of the hidden state $h_t$ is a hyper-parameter that affects the stored information memory capacity of the SSM within the Mamba layer. This section focuses on analyzing the effect of the hidden state dimension $h_t$ on hyperspectral tracking performance.

The impact of different hidden state dimensions is analyzed in Fig. 12. The results indicate that selecting an appropriate hidden state dimension is crucial for achieving optimal tracking performance. When $h_t$ is set to 16 in this article, CCTrack achieves the highest performance in terms of AUC and DP metrics. As $h_t$ increases progressively from 2 to 16 in powers of 2, both AUC and DP scores show steady improvement, indicating that a larger hidden state dimension enhances the model's ability to capture and store relevant information. However, when $h_t$ is set to 32, a decline in both evaluation metrics is observed, suggesting that overly large hidden state dimensions may introduce redundancy and instability.

In summary, balancing the hidden state dimension is essential for leveraging the full potential of the Mamba layer. While a higher dimension generally enhances the ability of CCTrack to model global contextual dependencies, an excessively large dimension can counteract these benefits by introducing noise and reducing the efficiency of the model. The choice of $h_t = 16$ in CCTrack strikes an effective balance, delivering the best performance while avoiding the drawbacks associated with higher dimensions.

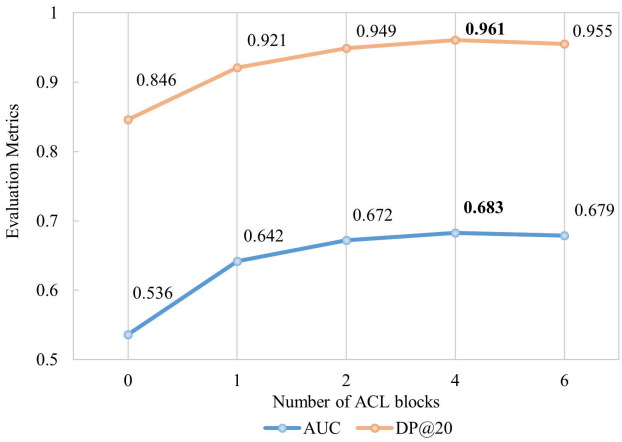



Fig. 11. Analysis experiments of the number of ACL blocks in CCTrack.

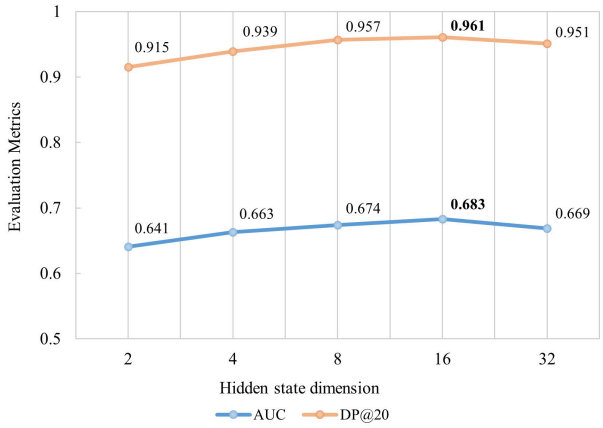



Fig. 12. Analysis experiments of hidden state dimension in the Mamba layer.

### I. Complexity Analysis

The Giga floating point operations (GFLOPs) and model parameters are reported in Table II to analyze the complexity of the proposed CCTrack in comparison with other advanced trackers. As shown in Table II, CCTrack achieves the lowest FLOPs and parameter count among transformer-based trackers such as SeqTrack and ARTrack. Specifically, the proposed

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Fig. 13. Visualizations of the context-aware representations in the proposed CCTrack.

TABLE II
COMPLEXITY ANALYSIS OF THE PROPOSED CCTRACK

| Tracker | FLOPs (G) | Parameters (M) |
|---|---|---|
| SiamRPN++ | **48.9** | **54** |
| SiamBAN | 51.5 | 53 |
| ARTrack | 514.7 | 458.5 |
| SeqTrack | 524 | 306.9 |
| **CCTrack** | 370.4 | 288.1 |

TABLE III
RUNNING SPEED OF THE PROPOSED CCTRACK

| | Methods | | | |
|---|---|---|---|---|
| | SeqTrack | ARTrack | HTACPE | **CCTrack** |
| Speed (FPS) | 5.7 | 5.2 | 3.82 | **14.1** |

method reduces FLOPs by 153.6G and parameters by 18.8 M compared to SeqTrack. Similarly, compared with ARTrack, CCTrack achieves a reduction of 143.6G in FLOPs and 170.4 M in parameters. This efficiency is primarily due to the absence of additional sequence modeling, which SeqTrack and ARTrack incorporate. These reductions emphasize the computational efficiency of CCTrack, allowing it to handle resources more effectively without compromising tracking performance. In contrast, although CCTrack has higher FLOPs and parameter counts compared to CNN-based trackers such as SiamRPN++ and SiamBAN, it significantly outperforms these methods in tracking accuracy. The limited feature modeling capacity and contextual understanding of CNN-based trackers restrict their ability to effectively handle complex scenarios, making CCTrack a more robust and reliable choice.

Overall, CCTrack achieves a promising balance between performance and model complexity. It achieves superior tracking results while maintaining a computational cost that is notably smaller than other transformer-based approaches, such as SeqTrack and ARTrack. The integration of context-aware learning and category consistency modeling enables CCTrack to optimize hyperspectral target tracking with minimal computational overhead. By effectively addressing the tradeoff between performance and complexity, CCTrack sets a new benchmark for efficient and robust tracking in the hyperspectral domain.

### J. Tracking Time Comparison

The tracking speed of the proposed CCTrack is compared with the state-of-art trackers, as shown in Table III. The results demonstrate that CCTrack achieves a significant improvement in tracking speed compared to other high-performance methods. Specifically, CCTrack is 3.69 times faster than HTACPE and outperforms SeqTrack and ARTrack by 2.47 and 2.71 times, respectively. These results highlight the efficiency of CCTrack in delivering high-speed tracking performance without sacrificing accuracy. The substantial speed advantage of CCTrack can be attributed to its lightweight design and efficient integration of context-aware learning and category consistency modeling. In contrast, SeqTrack and ARTrack exhibit higher computational costs due to their

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complex architectures. SeqTrack employs an encoder–decoder transformer, where the decoder autoregressively generates a sequence of bounding box values, leading to increased inference time. ARTrack, on the other hand, incorporates a sequence model that relies on previous target locations, which introduces additional processing overhead. Furthermore, both methods involve sequence modeling, making them less efficient for real-time tracking. By avoiding these costly operations and adopting an optimized structure, CCTrack effectively reduces computational overhead while maintaining robust tracking performance.

### K. Context-Aware Representations Analysis

The context-aware representations learned by CCTrack are visualized in Fig. 13, which illustrates the attention map of the final layer of the last ACL block. This visualization demonstrates how the proposed method focuses on the most relevant regions within hyperspectral scenes, effectively leveraging spectral and spatial information to enhance tracking accuracy. As shown in Fig. 13, in the "airplane1" and "boat" sequences, the targets are small in size and surrounded by significant background noise. Despite their small size, CCTrack accurately distinguishes the targets from the noisy background. In "car4," "double5," and "electriccar1" sequences, where distractors belong to the same class as the target, CCTrack effectively concentrates on the features of the correct target, exhibiting low attention on irrelevant objects. In the "doublecar3" sequence, however, which presents a challenging scenario with multiple objects of highly similar appearance, CCTrack successfully identifies subtle differences between the target and similar objects, maintaining high activation on the target while minimizing attention on distractors. The "doublecar8" represents a particular difficult situation, characterized by severe illumination darkness that disrupts normal tracking processes. Even in such adverse conditions, CCTrack accurately captures the representation of the target. Overall, the proposed CCTrack demonstrates superior performance across a range of challenging scenarios by learning context-aware representations in hyperspectral data.

## V. Conclusion

In this article, an HOT framework for remote sensing scenarios, CCTrack, is proposed to learn context-aware representations in hyperspectral data and establish interframe target consistency across video frames. The proposed CCTrack demonstrates strong capabilities in global information modeling and storage, while precisely localizing targets within a hyperspectral context. The tracker is evaluated on the remote sensing covered hyperspectral dataset IMEC25 and compared with state-of-the-art hyperspectral trackers. The results illustrate that the proposed tracker achieves superior performance on the benchmark, effectively capturing global contextual information and target category consistency for continuous and stable tracking. Future work will focus on developing a more efficient hyperspectral target tracking framework, with an emphasis on practical applications in engineering fields.

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